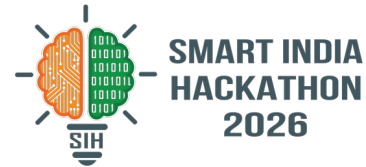
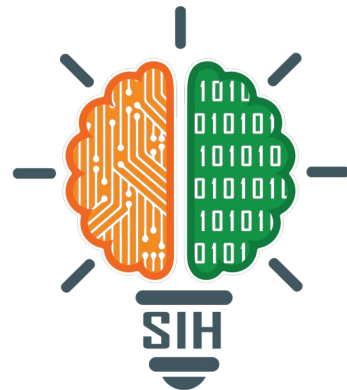


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – SIH26097
- Problem Statement Title – AI-driven voice assistant for livelihood mapping and NSQF-aligned skilling recommendations for SC communities under the GIA component of PM-AJAY
- Theme – Agriculture, FoodTech & Rural Development
- PS Category – Software
- Team ID – _____ (fill from the SIH portal)
- Team Name (Registered on portal) – Cognify



SMART INDIA HACKATHON 2026

Proposed solution · one free phone call that turns a person's skills into the right work



A form picks the course — not the person

so SC families end up in training that was never going to fit them

PM-AJAY pays up to ₹50,000 per SC family for skilling and small business. But ≈50,000 seats a year serve ≈5.9 crore eligible people — **fewer than 1 in 1,000**, so every seat has to fit.

41%

of trained people got a job

Source: CAG audit 2025

43%

of rural women have no phone

Source: NFHS-6, 2023-24

34%

of SC adults cannot read

Source: Census 2011

THE SOLUTION *one free call, about 10 minutes, on any keypad phone*



1

He gives a missed call

we call back, free

Within a minute, from a **govt 1600 number**, in the language he picks. **No app, no internet, no reading.**



2

Keys for facts, voice for skills

a speech error can never hurt

Age, education and distance are **pressed on keys**. Then he talks about his work — the AI maps it to an **official NCO code** and asks what he can already do.



3

1-4 options that actually fit

every fact checked before it is spoken

He hears his profile read back and presses 1 to confirm — **the AI writes it, no paper**. Then only courses, schemes or jobs near him. He can ask any doubt.



4

Apply, then follow-up

until he is actually earning

SMS with **Case ID + nearest CSC**; the officer approves in the AJAY app; SkillCall calls again at **1, 3 and 6 months**.

WHY IT WORKS WHERE A FORM FAILS *four things only a conversation can do*



Reaches him at all

any keypad phone, no reading

43% of rural women have no phone of their own and 34% of SC adults cannot read — **a form reaches neither**.



Asks like a person

each question follows his last answer

Bike mechanic → engine, tyres, oil — and it records the skills he is **still missing**.



Says only verified facts

govt data + SATYA double-check

Every answer comes from loaded rules, NQR courses and NCO codes; SATYA matches it to a row — **no match, not said**.



Writes the profile itself

officers only review and approve

The full case table is filled in the proper format, so **25 hours a month** of officer time comes back.

TECH STACK Indian, open where possible

Calls & voice

Exotel / Plivo

WhatsApp API

Pipecat voice server

AI models

Saaras v3 • STT

Bulbul v3 • TTS

Sarvam-30B / 105B

IndicConformer

Bhashini

SATYA (ours)

Backend & data

Python

FastAPI

Celery

Redis

PostgreSQL

pgvector

Apps & govt links

React

Leaflet maps

NCO / NQR

AJAY app*

DigiLocker*

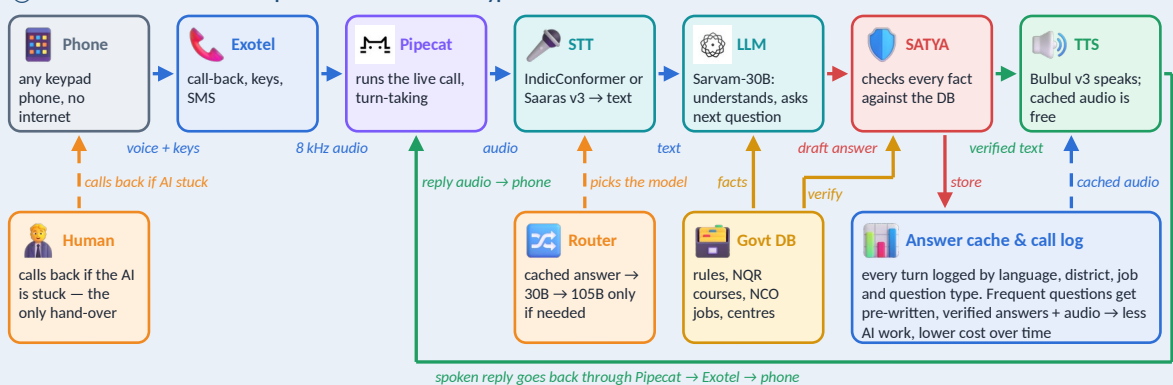
Docker • MeghRaj*

* needs a government MoU.

[Full architecture, API docs and code → GitHub repo \(add link\)](#)

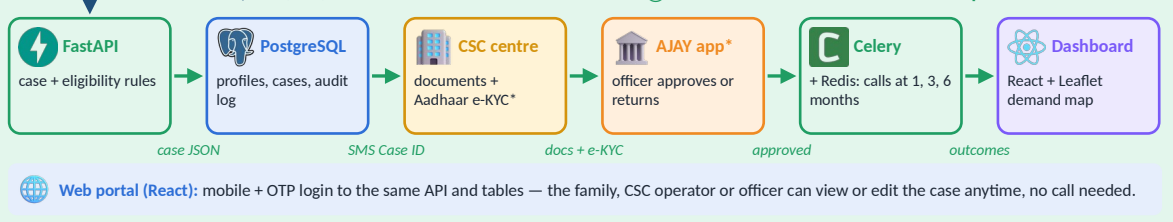
HOW IT WORKS two lanes: during the call (AI, in seconds) and after it (people, in days)

① DURING THE CALL — each spoken turn = 1.2-2.2 s • keypad turns are instant



call ends → confirmed profile (JSON) → FastAPI

② AFTER THE CALL — the confirmed profile becomes a case



AI understands → verified data decides → SATYA checks → humans approve

Data rules: no Aadhaar number stored · raw audio deleted after confirmation · consent logged (DPDP Rules 2025) · every edit in the audit log

Can it be built and run? **Yes** — every part already runs in India today, the prototype can be built now, and one completed call costs about ₹25–56.

ALREADY WORKING IN INDIA *nothing here is untested*



10 languages out of 22

are already in use

UIDAI runs Sarvam's voice AI for Aadhaar on its own servers.

Source: PIB / MeitY 2025



12 lakh calls a day

already handled by a govt line

Kilkari, the government health-advice line, runs this every day.

Source: BMJ Global Health 2021



≤ 20% word error

on real Indian phone calls

Best Indian speech AI, tested on 36,691 speakers in 15 languages.

Source: Voice of India 2026



1.2–2.2 s per reply

fast enough to feel like a talk

Measured end-to-end: speech in, verified answer spoken back.

Source: Sarvam docs · NVIDIA 2026

WHAT IT COSTS *four final numbers — every rate and step is in the calculations sheet*



₹25–56

for one completed call

A 16-minute call — interview, options and three follow-ups — at ≈ ₹1.6–3.5 a minute, all-in.



₹17,000–21,000

per 1,000 missed calls

420–517 people finish and get a verified profile; 40–100 of them need a human counsellor.



0.05–0.10%

of the PM-AJAY budget

₹1.1–2.1 crore a year covers all of India, against a budget of ₹2,140 crore.



Only 1% more

per person helped

The govt already spends ₹60,700 on one beneficiary; we add ₹220–420 so it goes to the right course.

₹9–13 lakh starts a 6-month pilot in one block (≈ 1,000 families) — and cost falls further as common answers get cached and speech moves to free self-hosted models. **Estimated** from Sarvam API pricing, Indian telephony rate cards 2026 and Kilkari (BMJ 2021); full working in the calculations sheet.

WHAT COULD GO WRONG — AND WHAT ALREADY HANDLES IT

✗ Village dialects are hard to understand ✓ facts by keypad, every answer read back, bigger model, then a human

✗ People ignore unknown numbers ✓ he calls first; the call-back comes from an official 1600 number

✗ The AI could say something wrong ✓ SATYA blocks anything not in the government database

✗ Slow replies bore callers ✓ streaming voice, pre-recorded questions, keypad turns are instant

Who gains? SC families, CSC operators, district officers and the Ministry. Every number below shows how it was worked out.

WHAT CHANGES *four numbers, each with its source*



41% → 70% placed

the gap we close

Only 41% of trained people get a job today; 70% is the government's own target. Per 1,000 trainees, even 50% means **90 more people working**.

Source: CAG 2025 · MSDE norms



62% → 73% stay

in the job they were placed in

+11 percentage points in a field trial, when a trainee is told the real pay and place before joining. SkillCall tells him on the call.

Source: J-PAL · Chakravorty et al. 2024



25 hours a month

of officer time comes back

Per officer handling 40 cases: **30 h → 5 h**. Across India that is **~ 18 officers' full-time work** freed every year.

Our estimate: AJAY app's data entry not counted



₹12.5–15 crore

a year, not wasted

If wrong matches fall in just **5 of every 100 cases** — that is **6–14×** what SkillCall costs to run.

Our estimate: PIB 2025 average support

1–1.5 lakh people profiled a year — 2–3 applicants screened for every GIA seat, so each of the $\approx 50,000$ seats goes to the person it fits best.

WHO GAINS WHAT *social · economic · governance · environment*



Social

SC families, women, non-readers

The right course the first time, free and in his own language — and calls until he is actually earning.



Economic

the Ministry and the trainee

Less money on wrong training; more people earning, **proved by follow-up** instead of assumed.



Governance

officers and CSC operators

One shared case record from call to job, plus a **demand map by block** for the yearly plan.



Environment

every district office

No paper forms, fewer trips to offices, and it runs on the **shared government cloud**.



Government data

PIB, Rajya Sabha reply (19 Mar 2025) — GIA: 2,01,006 beneficiaries, ₹1,219.80 cr pib.gov.in

PIB PM-AJAY explainer (May 2025) — eligibility, 5% admin & MIS line pib.gov.in

CAG Report No. 20 of 2025 — PMKVY: 41% placed, no skill-gap analysis cag.gov.in

NFHS-6 (2023-24) — women's phone ownership 63.6%, rural 57.4% rchiips.org

Census 2011 (MoSJE Handbook 2021) · SC poverty ratios 2011-12 (Lok Sabha) socialjustice.gov.in



Technology

Sarvam API pricing & docs — Saaras v3, Bulbul v3, 30B / 105B; voice-agent guide docs.sarvam.ai

NVIDIA × Sarvam 2026 — Sarvam-30B serving, first token < 1 s (P95) developer.nvidia.com

Artificial Analysis — Sarvam-30B ≈ \$0.03 per 1M tokens (blended) artificialanalysis.ai

AI4Bharat IndicConformer · Pipecat · Bhashini ai4bharat.iitm.ac.in



Research

Chakravorty et al. 2024 / J-PAL — honest job info: 62% → 73% stay in job povertyactionlab.org

Bashingwa et al. 2021, BMJ Global Health — Kilkari call answer rates gh.bmj.com

Voice of India 2026 — speech-AI benchmark on real Indian phone calls arxiv.org

Medhi et al. 2011, ACM TOCHI — voice interfaces for low-literacy users dl.acm.org



Standards & policy

PM-AJAY portal & AJAY app launch (26 May 2026) — PIB pib.gov.in

NCO-2015 occupation codes · NQR / NCVET job roles nqr.gov.in

DPDP Rules 2025 — consent and data protection meity.gov.in

TRAI 1600-series rules for government calls (2026) traf.gov.in

MeitY (Lok Sabha 2023) — 5.58 lakh CSCs, 4.44 lakh at gram panchayats eparlib.nic.in

SEE MORE *code, prototype, video and every calculation — one click each*



GitHub repo

code, architecture docs, API reference

[\(add link\) →](#)



Live prototype

try a SkillCall conversation in the browser

[\(add link\) →](#)



Demo video

2-minute walkthrough of a real call

[\(add link\) →](#)



Calculations sheet

every number, its source and full working

[\(add link\) →](#)